



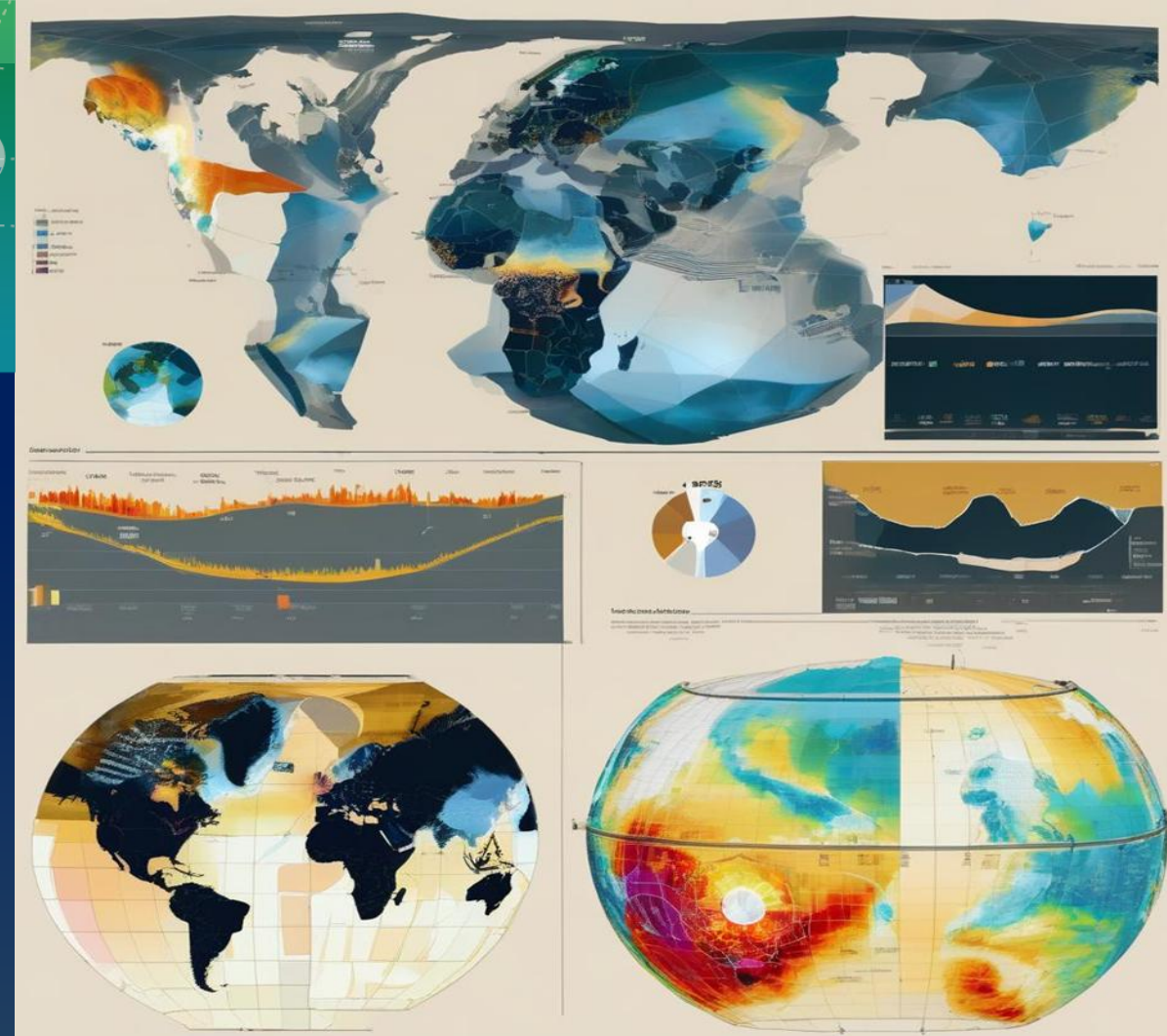
Food and Agriculture
Organization of the
United Nations



Building Capacity for Climate Risk Analysis: Methods, Tools, and Applications for Yemen's Agriculture and Related Sectors

Aden (Yemen), 2 Dec - 4 Dec., 2025

Session II. Ocean-atmospheric & climate analysis methodology



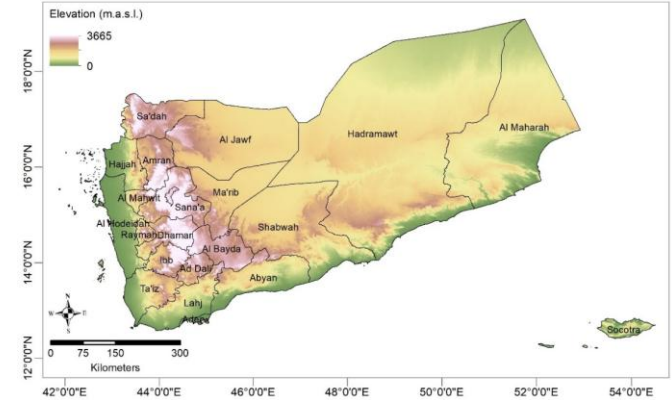
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Ocean-atmospheric factors affecting the Republic of Yemen

- Regional climate variability is largely determined by the large-scale atmospheric circulation (e.g. Intertropical Convergence Zone (ITCZ) and the Red Sea Convergence Zone (RSCZ)) ([Hael, 2021](#); [Farquharson et al. 1996](#)).
- The RSCZ leads to opposite prevailing winds, blowing from the northwest to the southeast at around 18-19°N from October to January, to those of the ITCZ.
- During the boreal summer months (July to September), the southwestern warm and humid winds carry water vapour from the Indian Ocean.
- The RSCZ and the ITCZ generate convective storms of high intensity and limited duration and extension, though the ITCZ storms have a larger spatial extent than those of the RSCZ ([Almhab and Tol, 2010](#)).



Ocean-atmospheric factors affecting the Republic of Yemen

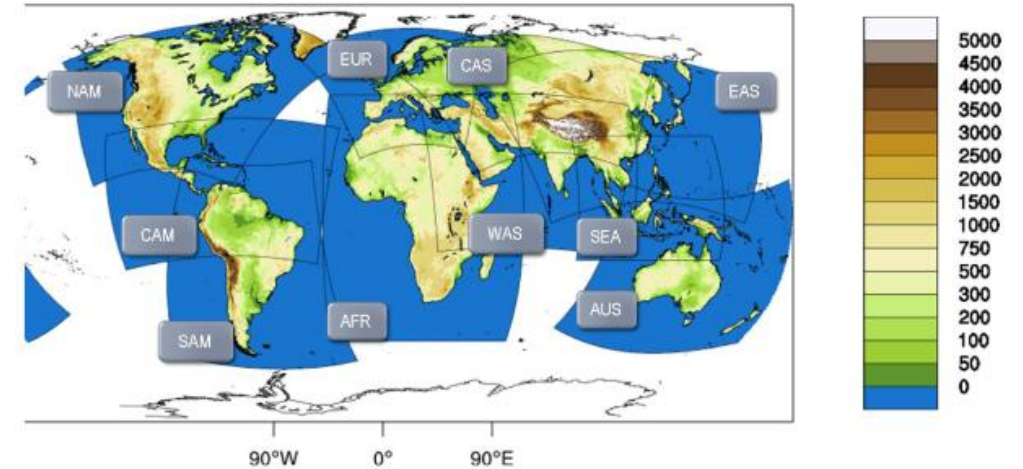
- During El Niño Southern Oscillation (ENSO), the RSCZ shifts towards the north, resulting in warmer sea surface temperatures (SST) and higher convective instability over the Red Sea. This phenomenon results in noticeable increases in the frequency and intensity of rainy days, and vice versa during La Niña years.
- The region is also affected by the Indian Ocean Dipole (IOD), characterized by contrasting SST between the south of the Arabian Peninsula and the eastern Indian Ocean.
- During a neutral IOD, cooler SST is observed over the western Indian Ocean, resulting in lower precipitation over the Arabic Peninsula, whereas a warmer than average western Indian Ocean (positive IOD) brings moisture to the Horn of Africa and the south of the Arabic Peninsula, including Yemen ([Dasari et al. 2018](#)).
- Statistically significant simultaneous teleconnections between IOD and ENSO have been reported in the literature ([Athar, 2015](#); [Chakraborty et al. 2006](#)), and its effects on summer monsoon variability depend upon the state of the phase and the amplitude of the two ([Charabi, 2009](#)).

Scientific approach CIPAs: historical climate (1990-2019)

- W5E5 is a gridded database (± 50 km) with global coverage. The Risk Team also has access to ERA5 dataset (± 25 km).
- W5E5 provides information for twelve climatic variables over the period 1980-2019 and is used as a reference in ISIMIP activities.
- This dataset (W5E5) is constructed from the ERA5 reanalysis, adjusted using the CRU TS4.03 and GPCCv2018 global observation datasets.
- **Climate variables used by the Risk Team:** daily Tmax ($^{\circ}\text{C}$), Tmin ($^{\circ}\text{C}$), PR (mm/%), RH (%), solar radiation (w/m^2), wind speed (m/s)

Scientific approach: future climate (2010-2100)

- CORDEX-CORE provides homogeneous regional climate projections for most inhabited land areas at a resolution of ± 25 km for the period 1970 to 2100.
- The Risk Team also has access to the CHELSA (± 1 km) dataset - though only for Small Island States



- Three general circulation models (GCMs) with high (HADGEM2-ES), medium (MPI-ESM), and low (NorESM) climate sensitivities are used in CIPAs. These models, representing the complete CMIP5 set, were selected to run regional simulations for two scenarios: RCP 2.6 and 8.5.
- The climate projections are dynamically downscaled using two regional climate models (REMO and RegCM4) and bias-corrected using historical data (W5E5) for the period 1976-2005.
- The choice of RCPS 2.6 and 8.5 covers the range of greenhouse gas concentration trajectories of the IPCC for the 21st century.

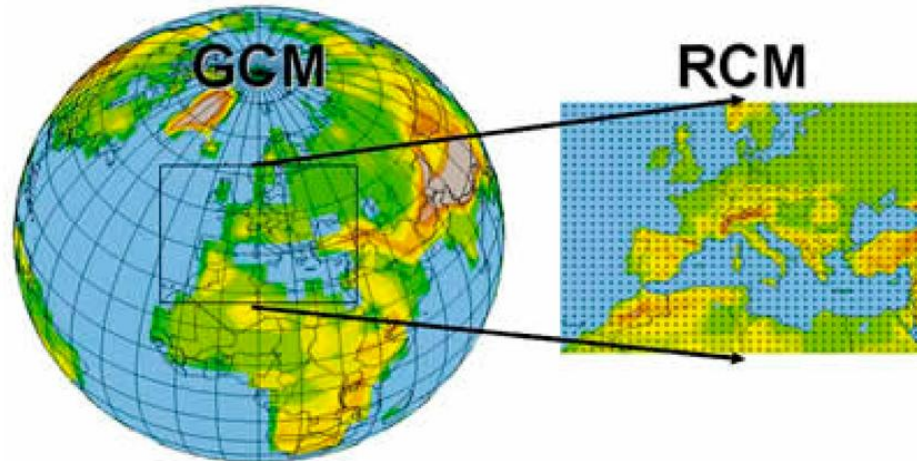
Scientific approach: climate analysis

Access to daily climatic data allows us to perform trend, variability and extreme analysis

Indicators	Code	Indicator name	Description	Units
Precipitation (PRCP)	PRCPTOT	Annual total wet-day PRCP	Annual total PRCP from wet days (days with PRCP > 1 mm)	mm
	DD	Number of dry days	Annual count of days with PRCP <1mm	Days
	CDD	Consecutive dry days	Consecutive dry days (maximum number of consecutive days with PRCP <1 mm)	Days
	R20	Number of heavy PRCP days	Annual count of days when PRCP > 20 mm	Days
	CWD	Consecutive wet days	Consecutive wet days (maximum number of consecutive days with PRCP > 1 mm)	Days
Temperature	SU	Summer days	Annual count of days with TX (daily maximum) > 25°C	Days
	TR	Tropical nights	Annual count of days with TN (daily minimum) > 20°C	Days
	HWVS	Heatwaves	Annual count of events with TX (daily maximum) > 35°C for 3 or more consecutive days	Events

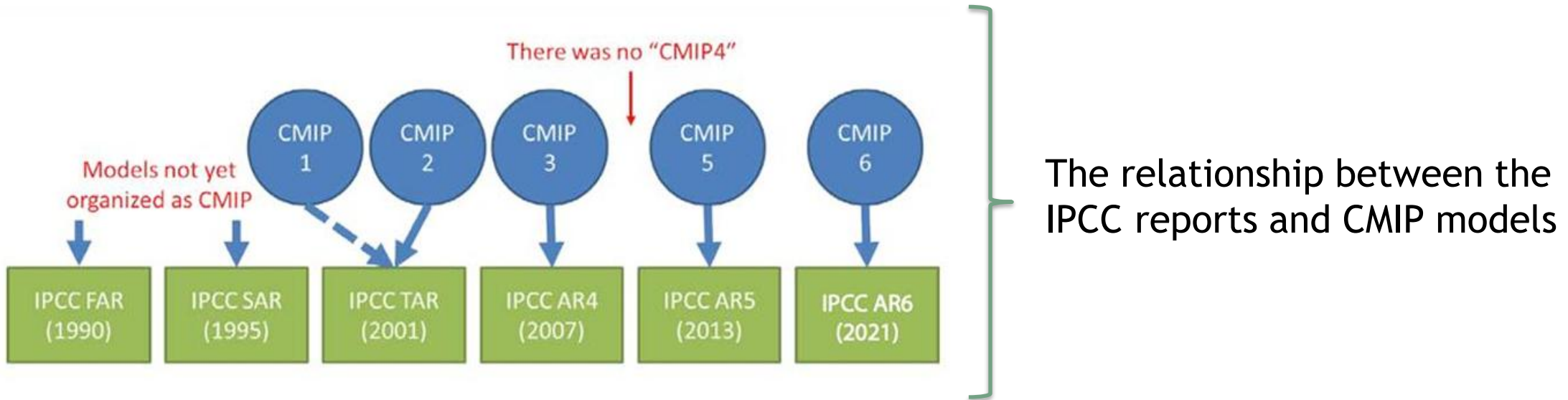
Scientific approach: GCMs vs. RCMs

Aspect	Global Climate Models (GCMs)	Regional Climate Models (RCMs)
Scale	•Global (entire earth)	•Régionale (concentres sur des zones spécifiques)
Grid size	•±100 à 500km or more	•±1 à 50 kilomètres
Implications	•Captures global climate trends, but not the local details	•Provides detailed simulations of local climatic processes
Use	•Long-term climate projections and global climate studies	•Regional climate and impact studies



Scientific approach: climate projections

- CORDEX-CORE models: These models potentially offer better resolution of the impacts of large-scale phenomena and spatial variability.
- ISIMIP CMIP6 data: This dataset is not dynamically refined, which could lead to a less accurate representation of local spatial precipitation models compared to CORDEX-CORE under these circumstances.



Scientific approach: climate projections

- The CMIP5 models consider different radiative forces (due to changes in CO₂, methane and aerosols) ranging from 2.6 to 8.5 W/m².

RCP	Relative radiative forcing	CO ₂ ppm	Global warming	CO ₂ emission pathway
1.9	1.9 W/m ²	450 ppm	1.5 °C	Zero by 2050
2.6	2.6 W/m ²	490 ppm	2 °C	Zero by 2075
4.5	4.5 W/m ²	650 ppm	3 °C	Falling after 2050
8.5	8.5 W/m ²	1370 ppm	5 °C	Keep rising

- Global warming levels (at the end of the century relative to 1800-1850) by different RCPs/SSPs

RCP (AR5)	SSP (AR6)
	SSP1-1.9 (≈ 1.4°C)
RCP2.6 (≈ 1.6°C)	SSP1-2.6 (≈ 1.8°C)
RCP4.5 (≈ 2.4°C)	SSP2-4.5 (≈ 2.7°C)
RCP6.0 (≈ 2.8°C)	SSP3-7.0 (≈ 3.6°C)
RCP8.5 (≈ 4.3°C)	SSP5-8.5 (≈ 4.4°C)

End!
